Shouvon Sarker

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Research Interests

Large Language Models (LLMs) • Neuro-Symbolic AI (Text-to-SQL) • Bayesian Deep Learning • Knowledge Distillation • Clinical Natural Language Processing • Trustworthy and Explainable AI • Counterfactual Reasoning

Education

Ph.D. in Electrical Engineering **Jan 2023 – Present**
Prairie View A&M University *Prairie View, TX, USA*

- **Dissertation (tentative):** *Enhancing Structured Predictions in Large Language Models*
- **Advisors:** Dr. Xishuang Dong, Dr. Lijun Qian

M.S. in Electrical Engineering **Aug 2021 – Dec 2022**
Prairie View A&M University *Prairie View, TX, USA*

- **Thesis:** *Medication Events Classification from Electronic Health Records Using BERT Models*
- **Advisors:** Dr. Xishuang Dong, Dr. Lijun Qian

B.S. in Electronics and Communication Engineering **Apr 2014 – Mar 2018**
Khulna University of Engineering and Technology *Khulna, Bangladesh*

Selected Graduate Coursework: Selected Topics in Deep Learning (Bayesian Networks, Variable Elimination, GANs); Modern Artificial Intelligence (Constraint Satisfaction Problems, Optimal Decisions in Games, Hidden Markov Models).

Publications

Manuscripts Under Review

1. **Sarker, S.**, Qian, L., & Dong, X. (2025). From Tokens to Transitions: A Structured Jensen–Shannon Knowledge Distillation Method for NER. *Under revision, IEEE Transactions on Knowledge and Data Engineering.*

2. **Sarker, S.**, Qian, L., & Dong, X. (2026). Inference-Time Bayesian Error Diagnosis and Repair for Text-to-SQL. *Submitted to Transactions of the Association for Computational Linguistics (TACL).*

3. **Sarker, S.**, Qian, L., & Dong, X. (2026). Task-Aware Circuit-Guided Knowledge Distillation for Interpretable Biomedical NER. *Submitted to Empirical Methods in Natural Language Processing (EMNLP).*

4. **Sarker, S.**, Qian, L., & Dong, X. (2026).Integrating LLMs-based Data Augmentation and Transformer Circuit to Enhance Performance and Interpretability in Medication Identification and Events Classification. *Submitted to International Conference on Intelligent Biology and Medicine (ICIBM)*

Peer-Reviewed Conference Proceedings

5. **Sarker, S.**, Qian, L., & Dong, X. (2025). Integrating Non-Parametric Attention to Enhance LLM-Based Text-to-SQL Without External Knowledge. *IEEE International Conference on Data Mining (ICDM).*
6. **Sarker, S.**, Qian, L., & Dong, X. (2025). Enhancing LLM Fine-tuning for Text-to-SQL by SQL Quality Measurement. *IEEE International Conference on Data Mining Workshops (ICDMW).*
7. **Sarker, S.**, Qian, L., & Dong, X. (2025). Text Generator and Text Discriminator for the NIST GenAI T2T Challenge. *International Conference on Artificial Intelligence, Robotics, and Control (AIRC).*
8. **Sarker, S.**, Li, X., & Dong, X. (2023). Medical Data Augmentation via ChatGPT: A Case Study on Medication Identification and Medication Event Classification. *IEEE–EMBS International Conference on Biomedical and Health Informatics (BHI).*
9. **Sarker, S.**, Qian, L., & Dong, X. (2023). Ensemble BERT for Medication Event Classification on Electronic Health Records. *International Conference on Intelligent Biology and Medicine (ICIBM).*
10. Orimoogunje, O., **Sarker, S.**, Li, X., & Qian, L. (2026). LLM-Assisted Task-Adaptive Semantic Communication over Non-Ideal Channels for Vision–Language Systems. *International Conference on Computer Communications and Networks (ICCCN).*
11. Kuo, M., **Sarker, S.**, Qian, L., et al. (2024). Enhancing Deep Knowledge Tracing via Diffusion Models for Personalized Adaptive Learning. *ASEE Annual Conference.*
12. Dong, X., **Sarker, S.**, & Qian, L. (2022). Integrating Human-in-the-Loop into Swarm Learning for Decentralized Fake News Detection. *International Conference on Data Science and Intelligent Applications (IDSTA).*

Poster Presentations

1. Improving LLM-Based Text-to-SQL Through Integrating Self-Discover Reasoning. *NASA DEAP Annual Meeting*, 2024.
2. Classification of Medication Events from Electronic Health Records Using BERT Models. *AI in Healthcare Symposium, Rice University*, 2024.

Research Experience

Graduate Research Assistant — **Jan 2023 – Present**
Prairie View A&M University, CREDIT Center — *Prairie View, TX*

- **Inference-Time Bayesian Refinement for Text-to-SQL:** Reformulated LLM-based SQL generation as an iterative loop of generation, probabilistic error diagnosis, and targeted repair.
- **Non-Parametric Attention:** Designed a schema-grounded attention mechanism that improves query generation accuracy by 10% without reliance on external knowledge bases.
- **Explainable Knowledge Distillation:** Proposed a Structured Jensen–Shannon divergence objective enabling interpretable transition-level supervision for Named Entity Recognition.
- **Riemannian Knowledge Distillation *(ongoing)*:** Extending student-model circuit projection onto the Stiefel manifold and enforcing Riemannian orthogonality for representational geometry.
- **Deployment:** Built and deployed an interactive Text-to-SQL system for querying scientific databases.

Team Lead — NIST GenAI Text-to-Text Challenge — **2024**
Prairie View A&M University — *Prairie View, TX*

- Designed a generative–critic framework for robustness evaluation of text generation systems.
- Achieved **Top-3 global placement** in the Generator Track (top 10% of participants).
- Invited speaker at the NIST GenAI T2T Workshop, 2024.

AI Assistant Developer — **Remote**
AMIE 2025 Conference — *2025*

- Developed an Android-based AI assistant using OpenAI GPT-4o and specific knowledge base.

Graduate Research Assistant — **Aug 2021 – Dec 2022**
Prairie View A&M University — *Prairie View, TX*

- Developed ensemble BERT architectures with calibration-aware prediction for clinical NLP, applied to medication event classification from electronic health records.

Teaching & Mentorship

Instructor — Prompt Engineering Workshop — **2024**
Prairie View A&M University — *Prairie View, TX*

- Delivered graduate-level instruction on Chain-of-Thought prompting, ReAct.

Research Mentor — **2023 – Present**
Prairie View A&M University — *Prairie View, TX*

- Supervised undergraduate ROTC student researchers on applied machine learning and NLP projects.

Professional Service

Conference & Journal Reviewer:

- IEEE World Congress on Computational Intelligence (WCCI 2026)
- IEEE International Joint Conference on Neural Networks (IJCNN 2025)
- IEEE International Conference on Data Mining (ICDM 2025)
- Springer Nature Computer Science

Awards & Honors

- **Top-3 Placement (Top 10%)**, NIST GenAI Text-to-Text Challenge, 2024.
- **Outstanding Student Award**, PVAMU CREDIT Center, 2024.
- **Top 10% Finalist**, n2c2 Clinical NLP Challenge, 2022.

Technical Skills

Programming Languages	Python (advanced), C/C++, SQL, HTML/CSS, PHP
Deep Learning Frameworks	PyTorch, TensorFlow, Keras, Hugging Face Transformers
LLM Tooling	LoRA, Adapters, PEFT, OpenAI API, LangChain
Data Science	Pandas, NumPy, scikit-learn, Matplotlib, Seaborn
Databases	MySQL, PostgreSQL, SQLite
Deployment & DevOps	Git, Docker, Linux, AWS

References

Available upon request.